MATH 39 — Linear Algebra

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Contents

1. Chapter 1: Systems of Linear Equations and Matrices

1.1 Linear Equations

Definition A **linear equation** in the variables $x_1, x_2, \ldots, x_n$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where $a_1, a_2, \ldots, a_n$ and b are constants. The constants $a_1, a_2, \ldots, a_n$ are called the **coefficients** of the equation, and b is called the **constant term**.

Examples of linear equations include:

- $2x + 3y - z = 5$
- $-x_1 + 4x_2 + 0x_3 = 7$
- $0x + 0y + 0z = 0$

Non-examples of linear equations include:

- $x^2 + y = 3$ (because of the x^2 term)
- $xy + z = 4$ (because of the xy term)
- $\sin(x) + y = 1$ (because of the $\sin(x)$ term)
- $x + \sqrt{y} = 2$ (because of the $\sqrt{y}$ term)

Systems of Linear Equations

Definition A **finite** collection of linear equations involving the same set of variables is called a **system of linear equations**.

Example of a system of linear equations:

$$\begin{cases} 2x + 3y - z = 5 \\ -x + 4y + 2z = 7 \\ 3x - y + z = 2 \\ \cdots \end{cases}$$

Definition A sequence of numbers $(s_1, s_2, \ldots, s_n)$ is called a **solution** to the system of linear equations if, when we substitute s_i for x_i in each equation, the equation holds true.

Example Consider the system of equations:

$$\begin{cases} x + y + z = 3 \\ 2x + y + 3z = 1 \end{cases}$$

Is the sequence $(-2, -5, 0)$ a solution to this system? Does $x = y = z = 1$?

Solution To check if $(-2, -5, 0)$ is a solution, substitute $x = -2$, $y = -5$, and $z = 0$ into both equations:

- First equation: $(-2) + (-5) + 0 = -7 \neq 3$ (does not hold)
- Second equation: $2(-2) + (-5) + 3(0) = -4 - 5 = -9 \neq 1$ (does not hold)

Thus, $(-2, -5, 0)$ is not a solution to the system.
Next, check if $(1, 1, 1)$ is a solution:

- First equation: $1 + 1 + 1 = 3$ (holds)
- Second equation: $2(1) + 1 + 3(1) = 2 + 1 + 3 = 6 \neq 1$ (does not hold)

Thus, $(1, 1, 1)$ is also not a solution to the system.

Solutions to Linear Systems

Definition There are three possible scenarios for the solutions to a system of linear equations:

- **No solution**: The system is inconsistent, meaning there are no values for the variables that satisfy all equations simultaneously.
- **Unique solution**: There is exactly one set of values for the variables that satisfies all equations.
- **Infinitely many solutions**: There are multiple sets of values for the variables that satisfy all equations. (i.e. the same line or scalar multiples of each other)

Systems of linear equations that have at least one solution are called **consistent**, while those with no solutions are called **inconsistent**.

Matricies

Definition A **matrix** is a rectangular array of numbers arranged in rows and columns. The numbers in the matrix are called its **entries** or **elements**. A matrix with m rows and n columns is called an $m \times n$ matrix.

Example of an **augmented** matrix:

$$\begin{cases} 3x_1 + 2x_2 - x_3 + x_4 = -1 \\ 2x_1 - x_3 + 2x_4 = 0 \\ 3x_1 + x_2 + 2x_3 + 5x_4 = 2 \end{cases} = \left[\begin{array}{cccc|c} 3 & 2 & -1 & 1 & -1 \\ 2 & 0 & -1 & 2 & 0 \\ 3 & 1 & 2 & 5 & 2 \end{array}\right]$$

Elementary Operations

Definition There are three types of **elementary row operations** that can be performed on a matrix:

- **Row swapping**: Interchanging two rows of the matrix.
- **Row scaling**: Multiplying all entries in a row by a non-zero constant.
- **Row addition**: Adding a multiple of one row to another row.

These operations are used in methods such as Gaussian elimination to solve systems of linear equations. Row operations do not produce equal matricies, but they do produce **equivalent** matrices, meaning they represent systems of equations with the same solution set. In order for 2 matricies to be equal, they must have the same dimensions and identical entries in corresponding positions.

Example Consider the matricies:

$$\left[\begin{array}{cc|c} 1 & 2 & 0 \\ 3 & 4 & 1 \end{array}\right] \mapsto \left[\begin{array}{cc|c} 2 & 4 & 0 \\ 3 & 4 & 1 \end{array}\right]$$

This represents the row operation of scaling the first row by 2 ($2R_1 \mapsto R_1$). These matricies are equivalent but not equal.

Example Solve:

$$\begin{cases} 2x+3y-z=3 \\ -x-y+z=1 \\ x-2y-5z=0 \end{cases} = \left[\begin{array}{ccc|c} 2 & 3 & -1 & 3 \\ -1 & -1 & 1 & 1 \\ 1 & -2 & -5 & 0 \end{array}\right] \xmapsto{R3 \mapsto R1} \left[\begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ -1 & -1 & 1 & 1 \\ 2 & 3 & -1 & 3 \end{array}\right] \xmapsto{R2+R1 \mapsto R2} \left[\begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & -3 & -4 & 1 \\ 2 & 3 & -1 & 3 \end{array}\right]$$

$$\xmapsto{R3-2R1 \mapsto R3} \left[\begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & -3 & -4 & 1 \\ 0 & 7 & 9 & 3 \end{array}\right] \xmapsto{-\frac{1}{3}R2 \mapsto R2} \left[\begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & 1 & \frac{4}{3} & -\frac{1}{3} \\ 0 & 7 & 9 & 3 \end{array}\right] \xmapsto{R3-7R2 \mapsto R3} \left[\begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & 1 & \frac{4}{3} & -\frac{1}{3} \\ 0 & 0 & \frac{-1}{3} & \frac{16}{3} \end{array}\right]$$

$$\xmapsto{-3R3 \mapsto R3} \left[\begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & 1 & \frac{4}{3} & -\frac{1}{3} \\ 0 & 0 & 1 & -16 \end{array}\right] \xmapsto{R2-\frac{4}{3}R3 \mapsto R2} \left[\begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & 1 & 0 & 21 \\ 0 & 0 & 1 & -16 \end{array}\right] \xmapsto{R1+5R3 \mapsto R1} \left[\begin{array}{ccc|c} 1 & -2 & 0 & -80 \\ 0 & 1 & 0 & 21 \\ 0 & 0 & 1 & -16 \end{array}\right]$$

$$\xmapsto{R1+2R2 \mapsto R1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -38 \\ 0 & 1 & 0 & 21 \\ 0 & 0 & 1 & -16 \end{array}\right] \implies (x,y,z) = (-38, 21, -16)$$

1.2 Gaussian Elimination

Gaussian elimination is a method for solving systems of linear equations by transforming the system's augmented matrix into a simpler form using elementary row operations. The goal is to reach **row echelon form** or **reduced row echelon form**, from which the solutions can be easily obtained.

Row Echelon Form

Definition A matrix is in **row echelon form** if it satisfies the following conditions:

- All non-zero rows are above any rows of all zeros.
- The leading entry (the first non-zero number from the left, also called a **pivot**) of each non-zero row is to the right of the leading entry of the previous row.
- The leading entry in any non-zero row is 1 (this condition is sometimes omitted in basic row echelon form).

Example of a matrix in row echelon form:

$$\begin{bmatrix} 1 & 2 & -1 & | & 3 \\ 0 & 1 & 3 & | & 4 \\ 0 & 0 & 1 & | & 2 \end{bmatrix}$$

It is often useful to further refine this to **reduced row echelon form**, which additionally requires that the leading 1 in each non-zero row is the only non-zero entry in its **column**.

Example of a matrix in reduced row echelon form:

$$\begin{bmatrix} 1 & -2 & 0 & | & 2 \\ 0 & 0 & 1 & | & -1 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Notes

- Every matrix can be reduced to row echelon form and reduced row echelon form using a sequence of elementary row operations.
- The reduced row echelon form of any matrix is unique.
- Row echelon form is not unique; different sequences of row operations can lead to different row echelon forms for the same matrix.

Zero Matrix A **zero matrix** is a matrix in which all entries are zero. It is often denoted by 0. The solution to the zero matrix is $\mathbb{R}^n$, where n is the number of variables.

Matrix Reduction Algorithm

1. Identify the leftmost non-zero column (the pivot column).
2. Select a non-zero entry in the pivot column as the pivot. If necessary, swap rows to move this entry to the top of the pivot column.
3. Scale the pivot row so that the pivot entry becomes 1.
4. Use row addition to create zeros in all positions below the pivot.
5. Cover the row containing the pivot and repeat the process for the submatrix that remains until all rows have been processed.
6. (Optional) To achieve reduced row echelon form, repeat the process to create zeros above each pivot as well.

Gaussian Elimination Algorithm

1. Augemnted Matrix -¿ RREF with Row Operations
2. If a row of the form [0 0 ... 0 — b] with b non-zero exists, then the system is inconsistent (no solutions).
3. Otherwise, assign free variables (if any) and solve for leading variables to find the solution set.
4. Write leading variables in terms of the parameters
5. If there are no nonleading variables, we gave 1 solution. If there are nonleading variables, we have infinitely many solutions.

Example 1 Solve the system:

$$\begin{cases} 3x + y - 4z = 1 \\ x + 10z = 5 \\ 4x + y + 6z = 1 \end{cases} = \left[\begin{array}{ccc|c} 3 & 1 & -4 & 1 \\ 1 & 0 & 10 & 5 \\ 4 & 1 & 6 & 1 \end{array}\right] \xmapsto{R2 \leftrightarrow R1} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 3 & 1 & -4 & 1 \\ 4 & 1 & 6 & 1 \end{array}\right] \xmapsto{R2 - 3R1 \mapsto R2} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -14 \\ 4 & 1 & 6 & 1 \end{array}\right]$$

$$\xmapsto{R3 - 4R1 \mapsto R3} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -14 \\ 0 & 1 & -34 & -19 \end{array}\right] \xmapsto{R3 - R2 \mapsto R3} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -14 \\ 0 & 0 & 0 & -5 \end{array}\right] \implies \text{no solutions}$$

Example 2 Solve the system:

$$\begin{cases} x_1 - 2x_2 - x_3 + 3x + 4 = 1 \\ 2x_1 - 4x_2 + x_3 = 5 \\ x_1 - 2x_2 + 2x_3 - 3x_4 = 4 \end{cases} = \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right] \xmapsto{R2-2R1\mapsto R2} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right]$$

$$\xmapsto{R3-R1\mapsto R3} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 3 & -6 & 3 \end{array}\right] \xmapsto{R3-R2\mapsto R3} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xmapsto{\frac{1}{3}R2\mapsto R2} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] = \begin{cases} x_1 - 2x_2 - x_3 + 3x_4 = 1 \\ x_3 - 2x_4 = 1 \end{cases}$$

$$\implies \begin{cases} x_2 = s \\ x_4 = t \\ x_1 = 1 + 2s + t \\ x_3 = 1 + 2t \end{cases}$$

Rank of a Matrix

Definition The **rank** of a matrix is the number of leading 1's (pivots) in its row echelon form or reduced row echelon form. It represents the dimension of the column space (or row space) of the matrix, which is the span of its columns (or rows). Rank is useful for finding the number of parameters in the solution set of a linear system.

Example 3 Consider the matrix from Example 2. Find the rank and number of parameters:

$$A = \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right] \xmapsto{\text{RREF(A)}} \left[\begin{array}{cccc|c} 1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \implies \text{rank}(A) = 2$$

$$n = 4 \text{ (number of variables)} \implies \text{number of parameters } = n - \text{rank}(A) = 4 - 2 = 2$$

Example 4 Find the rank of the following matricies:

$$A = \begin{bmatrix} 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \implies \text{rank}(A) = 2$$

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix} \xmapsto{R1\leftrightarrow R2} \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix} \xmapsto{\frac{1}{3}R1\mapsto R1} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \implies \text{rank}(B) = 1$$

$$C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \implies \text{rank}(C) = 0$$